

UNIVERSITY OF SOUTHAMPTON
Faculty of Engineering and Physical Sciences
School of Electronics and Computer Science

A progress report submitted for continuation towards a MEng

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**A Reproducible Evaluation of Linux
Rootkit Detectors via the Implementation
of an Automated Testbed &
Proof-of-Concept Rootkit**

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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
SCHOOL OF ELECTRONICS AND COMPUTER SCIENCE

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This work is all about ...

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Nomenclature

w The weight vector

Acknowledgements

Thanks to no one.

To ...

Chapter 1

Introduction

this will be the "enhanced description of your project"

1.1 Motivation

1.2 Project Objectives & Research Questions

1.3 Changes From Proposal

Chapter 2

Background & Related Work

2.1 Background Concepts

2.1.1 Linux & The Rootkit

Rootkits are typically described as a specialised piece of malware built for persistence & stealth which is installed *after* an attacker has gained root-level access to a system [1]. They may also give an attacker additional functionality such as network backdoors (cite magic bpf), the ability to hide files or other capabilities that give the attacker a greater ability to hide their presence whilst maintaining persistence.

Linux & subsequently Linux rootkits have been chosen for this project due to its relevance and prevalence in cloud & server environments. Additionally, a wide range of rootkit techniques exist for Linux which will be covered in the following section. Consequently, many tools have been developed which aim to detect the presence of a rootkit on a system. To achieve this, these tools may take a variety of approaches which will be covered in section 1.3.

The reliability and accuracy of these tools against current rootkits is paramount to ensure that test test...

2.1.2 Evolution of Rootkit Techniques

Linux rootkits have had an evolution

2.1.3 Rootkit Detectors

2.2 Related Work

narrow and positional, covers only the work closest to yours and exists to answer "how is this paper different?"

Chapter 3

Requirements

the part that explains the requirements of my engineering artefacts (the framework and mention the rootkit)

Chapter 4

Design & Methodology

the chapter that outlines the design plans for my engineering artefacts and also the methodology that will be used to investigate the RQ

4.1 rkbench

4.2 Rootkit Implementation

Chapter 5

Testing & Validation

The chapter that will go through my testing process for both the rootkit and the benchmarking system

Chapter 6

Progress To Date

An account of the current progress

Bibliography

- [1] National Institute of Standards and Technology, “Rootkit,” Computer Security Resource Center Glossary, n.d., accessed: 07-10-2026. [Online]. Available: <https://csrc.nist.gov/glossary/term/rootkit>